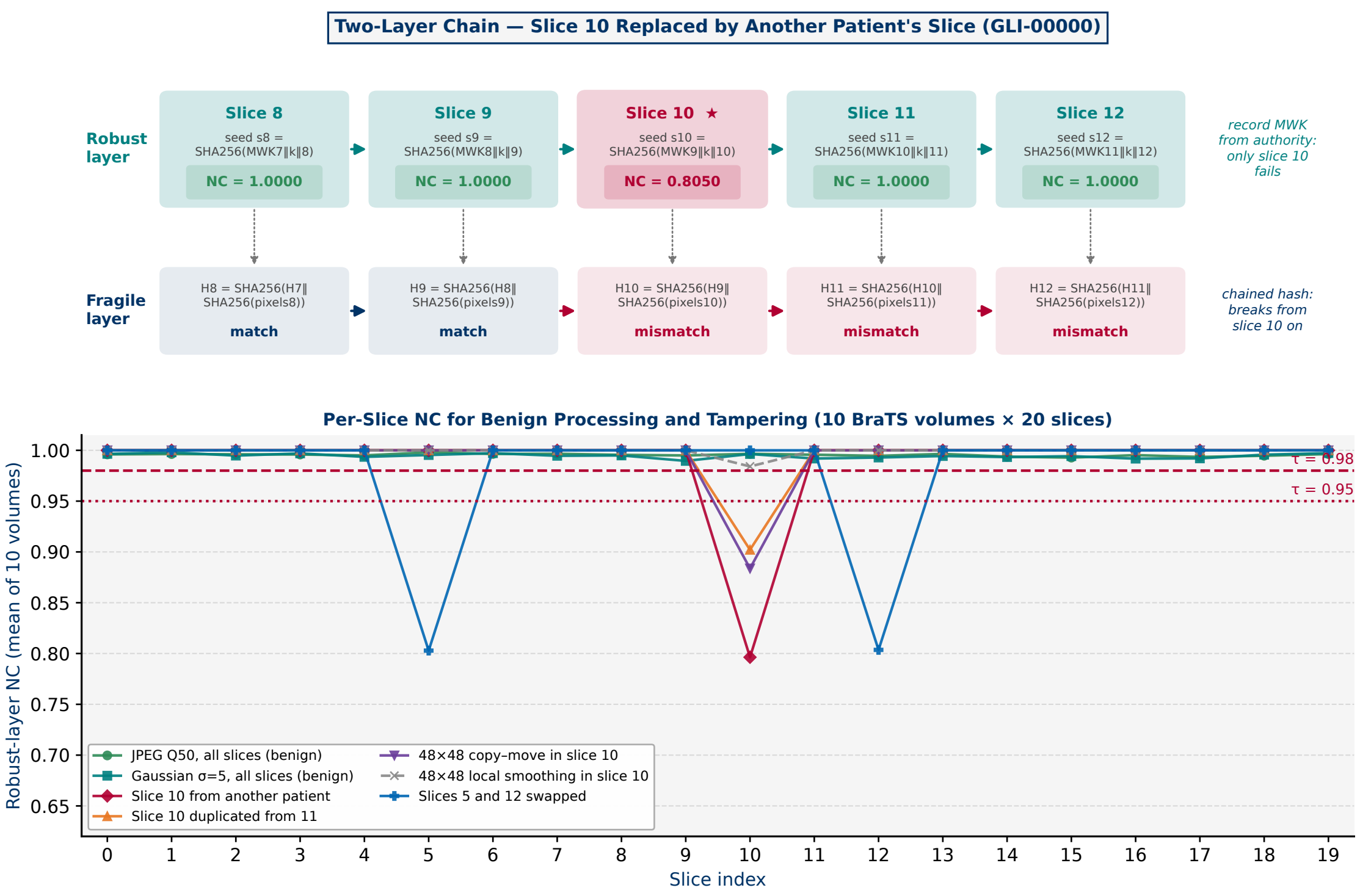
Two-Layer Chain — Slice 10 Replaced by Another Patient's Slice (GLI-00000)
Robust layer
Slice 8
seed s8 = SHA256(MWK7‖k‖8)
NC = 1.0000
Slice 9
seed s9 = SHA256(MWK8‖k‖9)
NC = 1.0000
Slice 10 ★
seed s10 = SHA256(MWK9‖k‖10)
NC = 0.8050
Slice 11
seed s11 = SHA256(MWK10‖k‖11)
NC = 1.0000
Slice 12
seed s12 = SHA256(MWK11‖k‖12)
NC = 1.0000
record MWK from authority: only slice 10 fails
Fragile layer
H8 = SHA256(H7‖ SHA256(pixels8))
match
H9 = SHA256(H8‖ SHA256(pixels9))
match
H10 = SHA256(H9‖ SHA256(pixels10))
mismatch
H11 = SHA256(H10‖ SHA256(pixels11))
mismatch
H12 = SHA256(H11‖ SHA256(pixels12))
mismatch
chained hash: breaks from slice 10 on
Per-Slice NC for Benign Processing and Tampering (10 BraTS volumes × 20 slices)
Robust-layer NC (mean of 10 volumes)
1.00
0.95
0.90
0.85
0.80
0.75
0.70
0.65
τ = 0.98
τ = 0.95
JPEG Q50, all slices (benign)
Gaussian σ=5, all slices (benign)
Slice 10 from another patient
Slice 10 duplicated from 11
48×48 copy–move in slice 10
48×48 local smoothing in slice 10
Slices 5 and 12 swapped
0
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
Slice index